The Impact of Cloud Computing and AI on Industry Dynamics and Competition

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Any opinions and conclusions expressed in this discussion do not necessarily represent the views of the Banque de France.

This paper: What it does

- Studies the effect of technological changes on industry dynamics
  - Firm entry
  - Firm exit ("true" exit and M&A)
  - Industry concentration
- Focuses on the adoption of two typical 2010's new technologies : cloud computing and AI
- Establishes 3 theoretical predictions based on the specificity of each technology & the existing literature
- Test empirically these predictions on Chinese firms

This paper: What it does

- Cloud & AI technological shock for the Chinese economy in 2013
- Builds an exposure index at the industry level based on two lists of keywords
  - Exposure = share of firms that have at least one of the keyword in their business scope description in 2012 $(t-1)$
- DID approach to measure the effects on different outcomes

This paper: What it finds

Main results - Following theoretical predictions

- Firm entry:
  - Cloud $+$
  - AI $\approx$
- Firm exit:
  - Cloud $+$ for all firms
  - AI $-$ for large firms, $+$ for small firms
- Industry concentration:
  - Cloud $-$
  - AI $+$

- Interpretation:
  - Incumbents not benefiting from cloud computing because it reduces entry costs and increases competition
  - Incumbents benefiting from AI because technology very data intensive

This paper: What it finds

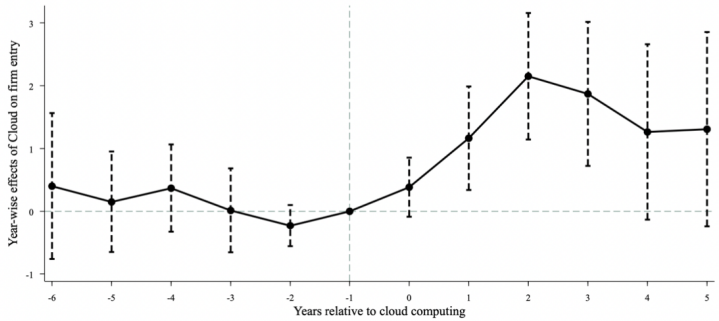
Additional results

- M&A:
  - Cloud + for all firms
  - AI + for large firms, − for small firms
- Equity / VC Equity:
  - Cloud +
  - AI +

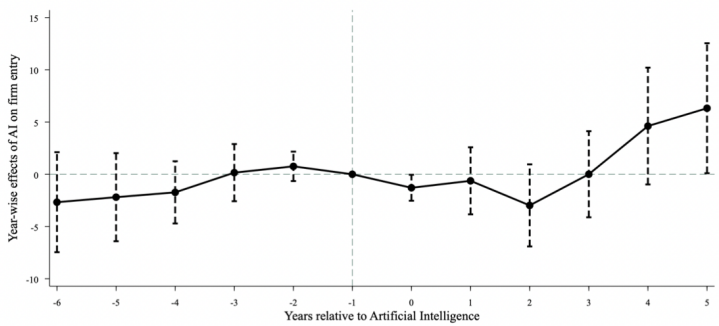
General Comment

- Great paper, which addresses a topical issue about the effect of technological adoption
- Provides evidence from Chinese firms
- Business dynamism approach : Great contribution to this literature
- Policy relevant:
  - GAFAM and the decline in business dynamism
  - Concentration issues with AI? (cf. European AI act)

Comment #1: 2013, a pivotal year



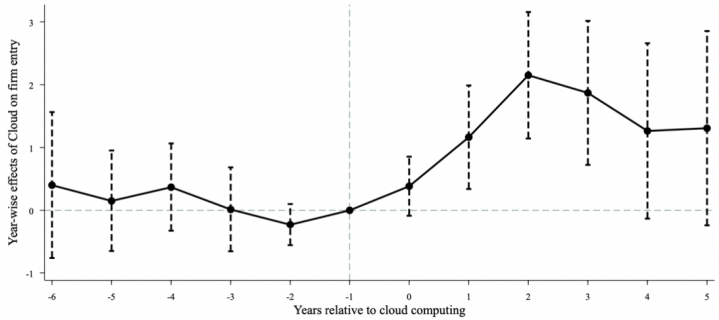
(a) Cloud Computing and Firm Entry



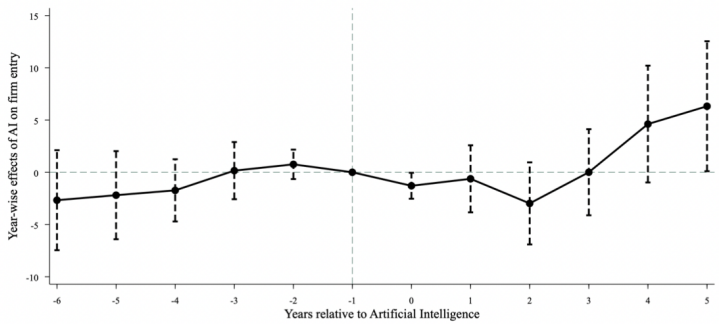
(b) AI and Firm Entry

- 2013 presented as the pivotal year for the adoption of AI and cloud computing in China
- Lack of information on the *effective* adoption year of cloud/AI technologies by firms within the industry
  $\longrightarrow$ DID provides reassuring pre-trend
  $\longrightarrow$ "Real" Event studies would be great: change in business scope description?

Comment #2: Causality



(a) Cloud Computing and Firm Entry



(b) AI and Firm Entry

- Possible confounding factor: Government intervention
- Following his AI development strategy, the Chinese government might have incentivized entry in some specific cloud/AI adopting industries after 2013 (public procurement, etc.)
- Possible instrument? Anecdotal evidence?

Comment #3: Heterogeneous effects?

- Point estimates for the effect of AI are quite large, larger than the effect of cloud computing, but very noisy
- Do we really have no effect of AI adoption on firm entry ...
- ... or heterogeneous effects between industries?

LEVEL & MODEL	Industry-Year & Poisson		
VARIABLES	Entry	Entry	Entry
	(1)	(2)	(3)
Post_Cloud	0.730	1.367*	0.944**
	(0.536)	(0.706)	(0.458)
Post_AI	2.286	2.562	0.593
	(2.493)	(2.598)	(2.115)
Constant	11.727***	12.518***	11.620***
	(0.022)	(0.785)	(1.159)
Observations	1,068	1,056	886
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Control×Year FE	No	Yes	No
L.Control	No	No	Yes

Broader Comment #1: Providers or users?

- Keyword selection:
  - Cloud
    - ★ Providers ("Hosting")
    - ★ Users ("online", "web", E-commerce")
  - AI
    - ★ Providers ("algorithms", "machine learning", "deep learning")
    - ★ Users ("automation", "robots")
- Heterogeneous effects between providers and adopters?
- Regarding AI, we could imagine higher fixed costs for providers (Extensive data) but lower fixed costs for adopters (AI enabling productivity gains for entrants)

Broader Comment #2: Can we really compare cloud and AI adoption?

- Cloud computing is more and more key for training of AI models
- For instance, the Villani report (2018) suggests the creation of a "cloud for AI" and "datacenter-as-a-service"
- The distinction seems relevant at the firm level but less at the industry level
- Possible to complement with firm level analysis?
- Add an interaction term between cloud and AI?